

Individual Mini-Project

Project Description

Select a critical non-signalized intersection within the Addis Ababa road network (use google earth). Conduct a comprehensive diagnosis to identify existing operational bottlenecks, geometric deficiencies, and safety hazards (e.g., conflict points, pedestrian interactions, or mixed-traffic challenges).

Using microscopic traffic simulation software (PTV Vissim or SUMO), develop a calibrated base model representing current conditions. Subsequently, design and simulate multiple improvement scenarios (e.g., geometric redesign, traffic calming measures, or conversion to signalized control). Finally, perform a comparative analysis of these scenarios using key performance indicators (KPIs) such as delay, queue length, throughput, and safety surrogate measures (e.g., SSAM conflict analysis) to propose an optimal solution.